

pandas2

September 24, 2025

```
[1]: import numpy as np
import pandas as pd
```

```
[2]: df = pd.read_csv('mckinsey.csv')
```

```
[6]: df.head(2)
```

```
[6]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030

```
[5]: df.tail(3)
```

```
[5]:
```

	country	year	population	continent	life_exp	gdp_cap
1701	Zimbabwe	1997	11404948	Africa	46.809	792.449960
1702	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1703	Zimbabwe	2007	12311143	Africa	43.487	469.709298

```
[8]: type(df['country'])
```

```
[8]: pandas.core.series.Series
```

```
[9]: se = df['country']
```

```
[12]: se[25]
```

```
[12]: 'Algeria'
```

```
[14]: # index wise element
```

```
[16]: # df[0]
```

```
[ ]:
```

```
[13]: # slicing
```

```
[17]: df[0:4]
```

```
[17]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138

```
[27]: df.index = list(range(1, len(df) + 1))
```

```
[28]: df
```

```
[28]:
```

	country	year	population	continent	life_exp	gdp_cap
1	Afghanistan	1952	8425333	Asia	28.801	779.445314
2	Afghanistan	1957	9240934	Asia	30.332	820.853030
3	Afghanistan	1962	10267083	Asia	31.997	853.100710
4	Afghanistan	1967	11537966	Asia	34.020	836.197138
5	Afghanistan	1972	13079460	Asia	36.088	739.981106
...	...	...	...	...	...	...
1700	Zimbabwe	1987	9216418	Africa	62.351	706.157306
1701	Zimbabwe	1992	10704340	Africa	60.377	693.420786
1702	Zimbabwe	1997	11404948	Africa	46.809	792.449960
1703	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1704	Zimbabwe	2007	12311143	Africa	43.487	469.709298

[1704 rows x 6 columns]

```
[ ]:
```

```
[30]: # Explicit indexing
df.loc[1]
```

```
[30]:
```

country	Afghanistan
year	1952
population	8425333
continent	Asia
life_exp	28.801
gdp_cap	779.445314

Name: 1, dtype: object

```
[31]: # Implicit indexing
df.iloc[0]
```

```
[31]:
```

country	Afghanistan
year	1952
population	8425333
continent	Asia
life_exp	28.801
gdp_cap	779.445314

Name: 1, dtype: object

```
[32]: df.iloc[-1]
```

```
[32]: country      Zimbabwe
      year        2007
      population  12311143
      continent   Africa
      life_exp    43.487
      gdp_cap     469.709298
      Name: 1704, dtype: object
```

```
[35]: len(df)
```

```
[35]: 1704
```

```
[34]: df.loc[len(df)]
```

```
[34]: country      Zimbabwe
      year        2007
      population  12311143
      continent   Africa
      life_exp    43.487
      gdp_cap     469.709298
      Name: 1704, dtype: object
```

```
[ ]:
```

```
[37]: # can i pass multiple indexes at the same time
```

```
[38]: df.head()
```

```
[38]:
```

	country	year	population	continent	life_exp	gdp_cap
1	Afghanistan	1952	8425333	Asia	28.801	779.445314
2	Afghanistan	1957	9240934	Asia	30.332	820.853030
3	Afghanistan	1962	10267083	Asia	31.997	853.100710
4	Afghanistan	1967	11537966	Asia	34.020	836.197138
5	Afghanistan	1972	13079460	Asia	36.088	739.981106

```
[41]: df.iloc[[1, 3, 5]]
```

```
[41]:
```

	country	year	population	continent	life_exp	gdp_cap
2	Afghanistan	1957	9240934	Asia	30.332	820.853030
4	Afghanistan	1967	11537966	Asia	34.020	836.197138
6	Afghanistan	1977	14880372	Asia	38.438	786.113360

```
[44]: df.loc[[1, 3, 5]]
```

```
[44]:
```

	country	year	population	continent	life_exp	gdp_cap
1	Afghanistan	1952	8425333	Asia	28.801	779.445314
3	Afghanistan	1962	10267083	Asia	31.997	853.100710
5	Afghanistan	1972	13079460	Asia	36.088	739.981106

```
[50]: df.reset_index(drop = True, inplace = True)
```

```
[51]: df
```

```
[51]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138
4	Afghanistan	1972	13079460	Asia	36.088	739.981106
...	...	...	...	...	...	...
1699	Zimbabwe	1987	9216418	Africa	62.351	706.157306
1700	Zimbabwe	1992	10704340	Africa	60.377	693.420786
1701	Zimbabwe	1997	11404948	Africa	46.809	792.449960
1702	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1703	Zimbabwe	2007	12311143	Africa	43.487	469.709298

[1704 rows x 6 columns]

```
[ ]:
```

```
[ ]:
```

0.0.1 Adding some data

```
[56]: df
```

```
[56]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138
4	Afghanistan	1972	13079460	Asia	36.088	739.981106
...	...	...	...	...	...	...
1699	Zimbabwe	1987	9216418	Africa	62.351	706.157306
1700	Zimbabwe	1992	10704340	Africa	60.377	693.420786
1701	Zimbabwe	1997	11404948	Africa	46.809	792.449960
1702	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1703	Zimbabwe	2007	12311143	Africa	43.487	469.709298

[1704 rows x 6 columns]

```
[ ]:
```

```
[57]: df.loc[1704] = ["India", 2025, 14000000000, 'Asia', 65, 900]
```

```
[59]: df.tail()
```

```
[59]:
```

	country	year	population	continent	life_exp	gdp_cap
1700	Zimbabwe	1992	10704340	Africa	60.377	693.420786
1701	Zimbabwe	1997	11404948	Africa	46.809	792.449960
1702	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1703	Zimbabwe	2007	12311143	Africa	43.487	469.709298
1704	India	2025	14000000000	Asia	65.000	900.000000

```
[60]: len(df)
```

```
[60]: 1705
```

```
[61]: df.loc[len(df)] = ["India", 2025, 14000000000, 'Asia', 65, 900]
```

```
[62]: df
```

```
[62]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138
4	Afghanistan	1972	13079460	Asia	36.088	739.981106
...	...	...	...	...	...	...
1701	Zimbabwe	1997	11404948	Africa	46.809	792.449960
1702	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1703	Zimbabwe	2007	12311143	Africa	43.487	469.709298
1704	India	2025	14000000000	Asia	65.000	900.000000
1705	India	2025	14000000000	Asia	65.000	900.000000

[1706 rows x 6 columns]

```
[64]: # df.iloc[len(df)] = ["India", 2025, 14000000000, 'Asia', 65, 900]
```

```
[65]: # If we want to insert a data/record we will have to use loc method
```

```
[68]: df.tail()
```

```
[68]:
```

	country	year	population	continent	life_exp	gdp_cap
1701	Zimbabwe	1997	11404948	Africa	46.809	792.449960
1702	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1703	Zimbabwe	2007	12311143	Africa	43.487	469.709298
1704	India	2025	14000000000	Asia	65.000	900.000000

```
1705      India  2022  14000000000      Asia   65.000  900.000000
```

```
[67]: df.iloc[1705] = ["India", 2022, 14000000000, 'Asia', 65, 900]
```

```
[69]: # Updation can be done using both loc and iloc
```

```
[71]: df.iloc[0]
```

```
[71]: country      Afghanistan
year              1952
population        8425333
continent         Asia
life_exp          28.801
gdp_cap           779.445314
Name: 0, dtype: object
```

```
[ ]:
```

```
[ ]:
```

```
[ ]:
```

Deleting data

```
[74]: df.loc[len(df.index)] = ['India', 2000, 13500000, 'Asia', 80.00, 900.23]
df.loc[len(df.index)] = ['India', 2000, 13500000, 'Asia', 80.00, 900.23]
df.loc[len(df.index)] = ['India', 2000, 13500000, 'Asia', 80.00, 900.23]
df.loc[len(df.index)] = ['India', 2000, 13500000, 'Asia', 80.00, 900.23]
```

```
[78]: df.tail(15)
```

```
[78]:
```

	country	year	population	continent	life_exp	gdp_cap
1702	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1703	Zimbabwe	2007	12311143	Africa	43.487	469.709298
1704	India	2025	14000000000	Asia	65.000	900.000000
1705	India	2022	14000000000	Asia	65.000	900.000000
1706	India	2000	13500000	Asia	80.000	900.230000
1707	India	2000	13500000	Asia	80.000	900.230000
1708	India	2000	13500000	Asia	80.000	900.230000
1709	India	2000	13500000	Asia	80.000	900.230000
1710	India	2000	13500000	Asia	80.000	900.230000
1711	India	2000	13500000	Asia	80.000	900.230000
1712	India	2000	13500000	Asia	80.000	900.230000
1713	India	2000	13500000	Asia	80.000	900.230000
1714	India	2000	13500000	Asia	80.000	900.230000
1715	India	2000	13500000	Asia	80.000	900.230000
1716	India	2000	13500000	Asia	80.000	900.230000

```
[85]: df.drop([1716, 1715, 1714])
```

```
[85]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138
4	Afghanistan	1972	13079460	Asia	36.088	739.981106
...	...	...	...	...	...	...
1709	India	2000	13500000	Asia	80.000	900.230000
1710	India	2000	13500000	Asia	80.000	900.230000
1711	India	2000	13500000	Asia	80.000	900.230000
1712	India	2000	13500000	Asia	80.000	900.230000
1713	India	2000	13500000	Asia	80.000	900.230000

[1714 rows x 6 columns]

```
[86]: # drop these columns inplace using drop method
```

```
[91]: # df.tail(15)
```

```
[88]: df.duplicated()
```

```
[88]:
```

0	False
1	False
2	False
3	False
4	False
...	...
1712	True
1713	True
1714	True
1715	True
1716	True

Length: 1717, dtype: bool

```
[92]: df[df.duplicated()]
```

```
[92]:
```

	country	year	population	continent	life_exp	gdp_cap
1707	India	2000	13500000	Asia	80.0	900.23
1708	India	2000	13500000	Asia	80.0	900.23
1709	India	2000	13500000	Asia	80.0	900.23
1710	India	2000	13500000	Asia	80.0	900.23
1711	India	2000	13500000	Asia	80.0	900.23
1712	India	2000	13500000	Asia	80.0	900.23
1713	India	2000	13500000	Asia	80.0	900.23
1714	India	2000	13500000	Asia	80.0	900.23

1715	India	2000	13500000	Asia	80.0	900.23
1716	India	2000	13500000	Asia	80.0	900.23

```
[94]: df.tail(15)
```

```
[94]:
```

	country	year	population	continent	life_exp	gdp_cap
1702	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1703	Zimbabwe	2007	12311143	Africa	43.487	469.709298
1704	India	2025	14000000000	Asia	65.000	900.000000
1705	India	2022	14000000000	Asia	65.000	900.000000
1706	India	2000	13500000	Asia	80.000	900.230000
1707	India	2000	13500000	Asia	80.000	900.230000
1708	India	2000	13500000	Asia	80.000	900.230000
1709	India	2000	13500000	Asia	80.000	900.230000
1710	India	2000	13500000	Asia	80.000	900.230000
1711	India	2000	13500000	Asia	80.000	900.230000
1712	India	2000	13500000	Asia	80.000	900.230000
1713	India	2000	13500000	Asia	80.000	900.230000
1714	India	2000	13500000	Asia	80.000	900.230000
1715	India	2000	13500000	Asia	80.000	900.230000
1716	India	2000	13500000	Asia	80.000	900.230000

```
[96]: df.drop_duplicates()
```

```
[96]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138
4	Afghanistan	1972	13079460	Asia	36.088	739.981106
...	...	...	...	...	...	...
1702	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1703	Zimbabwe	2007	12311143	Africa	43.487	469.709298
1704	India	2025	14000000000	Asia	65.000	900.000000
1705	India	2022	14000000000	Asia	65.000	900.000000
1706	India	2000	13500000	Asia	80.000	900.230000

[1707 rows x 6 columns]

```
[99]: df.drop_duplicates(keep = 'last')
```

```
[99]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138
4	Afghanistan	1972	13079460	Asia	36.088	739.981106

	...	...	...	...	...	...
1702	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1703	Zimbabwe	2007	12311143	Africa	43.487	469.709298
1704	India	2025	14000000000	Asia	65.000	900.000000
1705	India	2022	14000000000	Asia	65.000	900.000000
1716	India	2000	13500000	Asia	80.000	900.230000

[1707 rows x 6 columns]

```
[104]: df.drop_duplicates(keep = False, inplace = True)
```

```
[ ]:
```

```
[103]: df.duplicated(subset = ['country', 'year'])
```

```
[103]: 0      False
      1      False
      2      False
      3      False
      4      False
      ...
      1712    True
      1713    True
      1714    True
      1715    True
      1716    True
      Length: 1717, dtype: bool
```

```
[ ]:
```

```
[106]: # Quiz
```

```
[113]: data = {'A': ['foo', 'bar', 'foo', 'bar', 'foo', 'bar', 'foo', 'foo'],
               'B': ['one', 'one', 'two', 'three', 'two', 'two', 'one', 'three'],
               'C': ['small', 'large', 'large', 'small', 'small', 'large', 'large', 'small'],
               'D': [1, 2, 2, 3, 3, 4, 5, 6]}

DF = pd.DataFrame(data)

print(sum(DF.duplicated(subset=['A', 'B'])))
```

2

```
[111]: DF
```

```
[111]:      A      B      C  D
      0  foo   one  small  1
      1  bar   one  large  2
      2  foo   two  large  2
      3  bar  three  small  3
      4  foo   two  small  3
      5  bar   two  large  4
      6  foo   one  large  5
      7  foo  three  small  6
```

```
[115]: DF.duplicated(subset=['A', 'B'])
```

```
[115]: 0    False
      1    False
      2    False
      3    False
      4     True
      5    False
      6     True
      7    False
      dtype: bool
```

```
[ ]:
```

```
[ ]:
```

Slicing

- iloc and loc revisited

```
[116]: # Get values from row 0 to 4 both inclusive
```

```
[117]: df
```

```
[117]:      country  year  population  continent  life_exp  gdp_cap
      0  Afghanistan  1952      8425333      Asia    28.801  779.445314
      1  Afghanistan  1957      9240934      Asia    30.332  820.853030
      2  Afghanistan  1962     10267083      Asia    31.997  853.100710
      3  Afghanistan  1967     11537966      Asia    34.020  836.197138
      4  Afghanistan  1972     13079460      Asia    36.088  739.981106
      ...      ...      ...
      1701  Zimbabwe  1997     11404948  Africa    46.809  792.449960
      1702  Zimbabwe  2002     11926563  Africa    39.989  672.038623
      1703  Zimbabwe  2007     12311143  Africa    43.487  469.709298
      1704      India  2025  14000000000      Asia    65.000  900.000000
      1705      India  2022  14000000000      Asia    65.000  900.000000
```

[1706 rows x 6 columns]

```
[118]: df.iloc[0:4]
```

```
[118]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138

```
[119]: # End is exclusive in iloc:
```

```
df.iloc[0:4, 0:3]
```

```
[119]:
```

	country	year	population
0	Afghanistan	1952	8425333
1	Afghanistan	1957	9240934
2	Afghanistan	1962	10267083
3	Afghanistan	1967	11537966

```
[121]: # In loc -> end is also inclusive
df.loc[0:4]
```

```
[121]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138
4	Afghanistan	1972	13079460	Asia	36.088	739.981106

```
[122]: # using loc
```

```
[124]: df.head()
```

```
[124]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138
4	Afghanistan	1972	13079460	Asia	36.088	739.981106

```
[129]: # This is not going to work
# df.iloc[0:4, 0:3]
```

```
[130]: df.loc[0:4, ['country', 'year', 'population']]
```

```
[130]:      country  year  population
0  Afghanistan  1952    8425333
1  Afghanistan  1957    9240934
2  Afghanistan  1962   10267083
3  Afghanistan  1967   11537966
4  Afghanistan  1972   13079460
```

```
[140]: df.head()
```

```
[140]:      country  year  population  continent  life_exp  gdp_cap
0  Afghanistan  1952    8425333        Asia    28.801  779.445314
1  Afghanistan  1957    9240934        Asia    30.332  820.853030
2  Afghanistan  1962   10267083        Asia    31.997  853.100710
3  Afghanistan  1967   11537966        Asia    34.020  836.197138
4  Afghanistan  1972   13079460        Asia    36.088  739.981106
```

```
[133]: df.loc[0:5, 'country' : 'population']
```

```
[133]:      country  year  population
0  Afghanistan  1952    8425333
1  Afghanistan  1957    9240934
2  Afghanistan  1962   10267083
3  Afghanistan  1967   11537966
4  Afghanistan  1972   13079460
5  Afghanistan  1977   14880372
```

```
[ ]:
```

```
[136]: # HW:
# df.iloc[[0,10,100], [0,2,3]]
# df.loc[0:10:2, 'country'::2]
```

```
[135]: df.loc[0:10:2]
```

```
[135]:      country  year  population  continent  life_exp  gdp_cap
0  Afghanistan  1952    8425333        Asia    28.801  779.445314
2  Afghanistan  1962   10267083        Asia    31.997  853.100710
4  Afghanistan  1972   13079460        Asia    36.088  739.981106
6  Afghanistan  1982   12881816        Asia    39.854  978.011439
8  Afghanistan  1992   16317921        Asia    41.674  649.341395
10 Afghanistan  2002   25268405        Asia    42.129  726.734055
```

```
[ ]:
```

```
[ ]:
```

Pandas Inbuilt Operations

- Aggregate function

```
[141]: df.head()
```

```
[141]:      country  year  population  continent  life_exp  gdp_cap
0  Afghanistan  1952    8425333        Asia    28.801  779.445314
1  Afghanistan  1957    9240934        Asia    30.332  820.853030
2  Afghanistan  1962   10267083        Asia    31.997  853.100710
3  Afghanistan  1967   11537966        Asia    34.020  836.197138
4  Afghanistan  1972   13079460        Asia    36.088  739.981106
```

```
[143]: le = df['life_exp']
```

```
[144]: # find avg/mean life expectancy
```

```
[145]: le.mean()
```

```
[145]: 59.480917162954235
```

```
[146]: le.sum()
```

```
[146]: 101474.44467999999
```

```
[147]: sum(le)
```

```
[147]: 101474.44467999993
```

```
[148]: le.min()
```

```
[148]: 23.599
```

```
[149]: le.max()
```

```
[149]: 82.603
```

```
[ ]:
```

```
[150]: # Find country with highest gdp_cap
```

```
[153]: max_gdp = df['gdp_cap'].max()
```

```
[155]: df[df['gdp_cap'] == max_gdp]
```

```
[155]:      country  year  population  continent  life_exp  gdp_cap
853  Kuwait    1957    212846        Asia    58.033  113523.1329
```

```
[156]: df[df['gdp_cap'] == max_gdp].index
```

```
[156]: Int64Index([853], dtype='int64')
```

```
[ ]:
```

```
[ ]:
```

0.0.2 Sorting

```
[157]: df
```

```
[157]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138
4	Afghanistan	1972	13079460	Asia	36.088	739.981106
...	...	...	...	...	...	...
1701	Zimbabwe	1997	11404948	Africa	46.809	792.449960
1702	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1703	Zimbabwe	2007	12311143	Africa	43.487	469.709298
1704	India	2025	14000000000	Asia	65.000	900.000000
1705	India	2022	14000000000	Asia	65.000	900.000000

[1706 rows x 6 columns]

```
[159]: df.sort_values(by = 'life_exp')
```

```
[159]:
```

	country	year	population	continent	life_exp	gdp_cap
1292	Rwanda	1992	7290203	Africa	23.599	737.068595
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
552	Gambia	1952	284320	Africa	30.000	485.230659
36	Angola	1952	4232095	Africa	30.015	3520.610273
1344	Sierra Leone	1952	2143249	Africa	30.331	879.787736
...	...	...	...	...	...	...
1487	Switzerland	2007	7554661	Europe	81.701	37506.419070
695	Iceland	2007	301931	Europe	81.757	36180.789190
802	Japan	2002	127065841	Asia	82.000	28604.591900
671	Hong Kong, China	2007	6980412	Asia	82.208	39724.978670
803	Japan	2007	127467972	Asia	82.603	31656.068060

[1706 rows x 6 columns]

```
[160]: df.sort_values(['life_exp'])
```

```
[160]:
```

	country	year	population	continent	life_exp	gdp_cap
1292	Rwanda	1992	7290203	Africa	23.599	737.068595
0	Afghanistan	1952	8425333	Asia	28.801	779.445314

552	Gambia	1952	284320	Africa	30.000	485.230659
36	Angola	1952	4232095	Africa	30.015	3520.610273
1344	Sierra Leone	1952	2143249	Africa	30.331	879.787736
...	...	...	...	...	...	...
1487	Switzerland	2007	7554661	Europe	81.701	37506.419070
695	Iceland	2007	301931	Europe	81.757	36180.789190
802	Japan	2002	127065841	Asia	82.000	28604.591900
671	Hong Kong, China	2007	6980412	Asia	82.208	39724.978670
803	Japan	2007	127467972	Asia	82.603	31656.068060

[1706 rows x 6 columns]

```
[162]: df.sort_values(['life_exp'], ascending=False)
```

```
[162]:
```

	country	year	population	continent	life_exp	gdp_cap
803	Japan	2007	127467972	Asia	82.603	31656.068060
671	Hong Kong, China	2007	6980412	Asia	82.208	39724.978670
802	Japan	2002	127065841	Asia	82.000	28604.591900
695	Iceland	2007	301931	Europe	81.757	36180.789190
1487	Switzerland	2007	7554661	Europe	81.701	37506.419070
...	...	...	...	...	...	...
1344	Sierra Leone	1952	2143249	Africa	30.331	879.787736
36	Angola	1952	4232095	Africa	30.015	3520.610273
552	Gambia	1952	284320	Africa	30.000	485.230659
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1292	Rwanda	1992	7290203	Africa	23.599	737.068595

[1706 rows x 6 columns]

```
[164]: df.sort_values(['year', 'life_exp'])
```

```
[164]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
552	Gambia	1952	284320	Africa	30.000	485.230659
36	Angola	1952	4232095	Africa	30.015	3520.610273
1344	Sierra Leone	1952	2143249	Africa	30.331	879.787736
1032	Mozambique	1952	6446316	Africa	31.286	468.526038
...	...	...	...	...	...	...
695	Iceland	2007	301931	Europe	81.757	36180.789190
671	Hong Kong, China	2007	6980412	Asia	82.208	39724.978670
803	Japan	2007	127467972	Asia	82.603	31656.068060
1705	India	2022	14000000000	Asia	65.000	900.000000
1704	India	2025	14000000000	Asia	65.000	900.000000

[1706 rows x 6 columns]

```
[165]: df.sort_values(['year', 'life_exp'], ascending=[False, True])
```

```
[165]:
```

	country	year	population	continent	life_exp	gdp_cap
1704	India	2025	14000000000	Asia	65.000	900.000000
1705	India	2022	14000000000	Asia	65.000	900.000000
1463	Swaziland	2007	1133066	Africa	39.613	4513.480643
1043	Mozambique	2007	19951656	Africa	42.082	823.685621
1691	Zambia	2007	11746035	Africa	42.384	1271.211593
...	...	...	...	...	...	...
408	Denmark	1952	4334000	Europe	70.780	9692.385245
1464	Sweden	1952	7124673	Europe	71.860	8527.844662
1080	Netherlands	1952	10381988	Europe	72.130	8941.571858
684	Iceland	1952	147962	Europe	72.490	7267.688428
1140	Norway	1952	3327728	Europe	72.670	10095.421720

[1706 rows x 6 columns]

```
[ ]:
```

```
[166]: # Quiz:
```

```
[167]: # df.sort_values(['country', 'population'])

# df.sort_values(['country', 'population'], inplace=True)

# df.sort_values(['country', 'population'], inplace=True, ascending=False) ->
↳ correct

# df.sort_values(['country', 'population'], inplace=True, ascending=True)
```

```
[ ]:
```

```
[ ]:
```

```
[ ]:
```

Concatenation and Merging

```
[169]: users = pd.DataFrame({"userid": [1, 2, 3], "name": ["Suvan", "Rahul",
↳ "khusalli"]})

users
```

```
[169]:
```

	userid	name
0	1	Suvan
1	2	Rahul
2	3	khusalli

```
[170]: msgs = pd.DataFrame({"userid": [1, 1, 2, 4], "msg": ['hmm', "acha", "theek hai",
↳ "nice"]})
```

```
msgs
```

```
[170]:
```

	userid	msg
0	1	hmm
1	1	acha
2	2	theek hai
3	4	nice

```
[ ]:
```

```
[171]: # Concatenation
```

```
[172]: # concat using axis = 0
```

```
[174]: # pd.concat?
```

```
[175]: pd.concat([users, msgs])
```

```
[175]:
```

	userid	name	msg
0	1	Suvan	NaN
1	2	Rahul	NaN
2	3	khusalli	NaN
0	1	NaN	hmm
1	1	NaN	acha
2	2	NaN	theek hai
3	4	NaN	nice

```
[176]: pd.concat([users, msgs], axis = 1)
```

```
[176]:
```

	userid	name	userid	msg
0	1.0	Suvan	1	hmm
1	2.0	Rahul	1	acha
2	3.0	khusalli	2	theek hai
3	NaN	NaN	4	nice

```
[ ]:
```

```
[ ]:
```

```
[177]: # merge
```

```
[180]: users
```

```
[180]:
```

	userid	name
0	1	Suvan
1	2	Rahul
2	3	khusalli

```
[181]: msgs
```

```
[181]:   userid      msg
0      1      hmm
1      1      acha
2      2  theek hai
3      4      nice
```

```
[179]: pd.merge(users, msgs, on='userid')
```

```
[179]:   userid  name      msg
0      1  Suvan      hmm
1      1  Suvan      acha
2      2  Rahul  theek hai
```

```
[182]: users.merge(msgs, on='userid')
```

```
[182]:   userid  name      msg
0      1  Suvan      hmm
1      1  Suvan      acha
2      2  Rahul  theek hai
```

```
[185]: pd.merge(users, msgs, on='userid', how='left')
```

```
[185]:   userid  name      msg
0      1  Suvan      hmm
1      1  Suvan      acha
2      2  Rahul  theek hai
3      3 khusalli      NaN
```

```
[186]: pd.merge(users, msgs, on='userid', how='right')
```

```
[186]:   userid  name      msg
0      1  Suvan      hmm
1      1  Suvan      acha
2      2  Rahul  theek hai
3      4   NaN      nice
```

```
[188]: # HW: outer and cross joins
```

```
[ ]:
```

```
[192]: # Following is a self join
```

```
[193]: pd.merge(users, users, on='userid')
```

```
[193]:
```

	userid	name_x	name_y
0	1	Suvan	Suvan
1	2	Rahul	Rahul
2	3	khusalli	khusalli

```
[ ]:
```

```
[194]: # Quiz
```

```
[195]: df1 = pd.DataFrame({'A':[10,30], 'B':[20,40], 'C':[30, 60]})  
df2 = pd.DataFrame({'A':[10,30], 'C':[30, 60]})  
df2.merge(df1, on = 'A', how = 'outer')
```

```
[195]:
```

	A	C_x	B	C_y
0	10	30	20	30
1	30	60	40	60

```
[196]: df2.merge(df1, on = 'A', how = 'outer').shape
```

```
[196]: (2, 4)
```

```
[ ]:
```

```
[ ]:
```

```
[ ]:
```

```
[ ]:
```